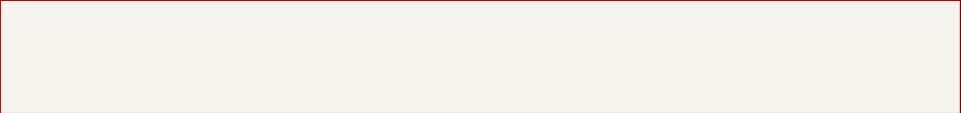


MACH3-SIMPLE – 6 axes, two LPT ports, buffered 5 V outputs

A copy of the board in service, extras removed. NOT ISOLATED on the output side.

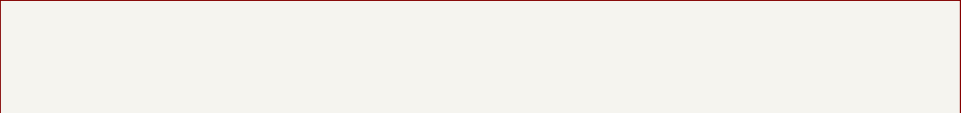
A1: NOT BUILT, NOT TESTED. 24 V in only; isolated VFD port.

Power – 24 V in, 5 V made on board



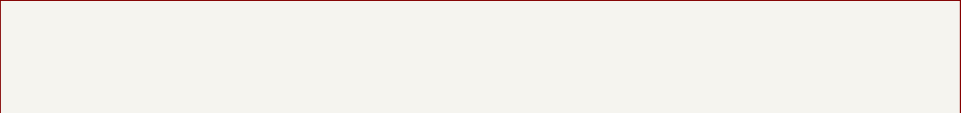
File: 01_power.kicad_sch

Inputs – home, E-stop, THC, handwheel



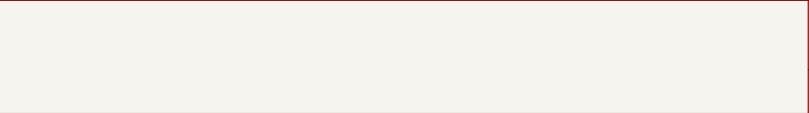
File: 03_inputs.kicad_sch

Spindle – isolated 0–10 V and FWD–REV for the VFD



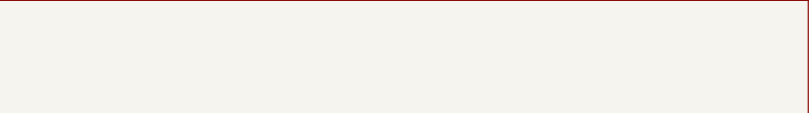
File: 05_spindle.kicad_sch

LPT port 1 – 6 axes, buffered 5 V out



File: 02_outputs.kicad_sch

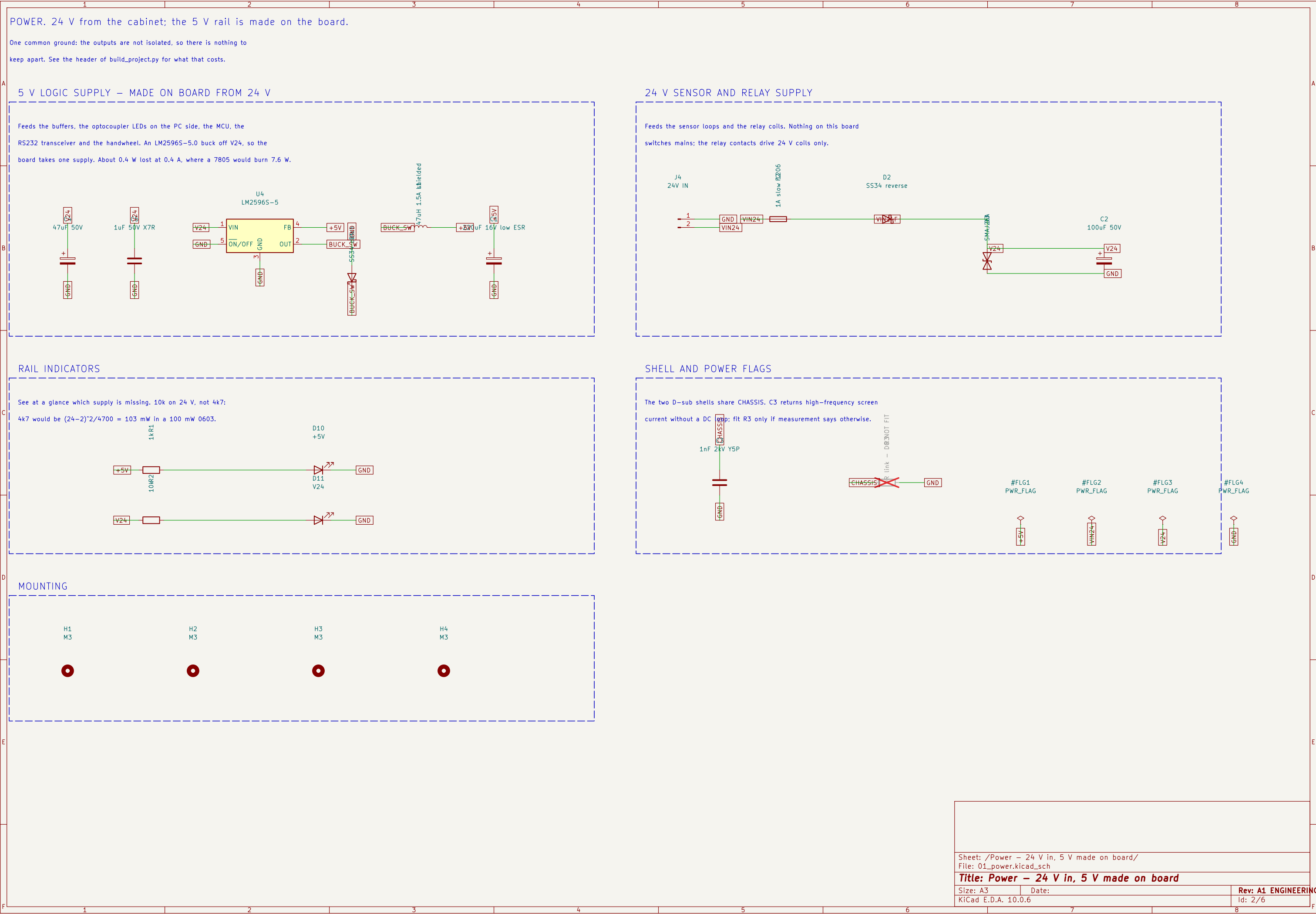
RS232, MCU and six relays



File: 04_relays.kicad_sch

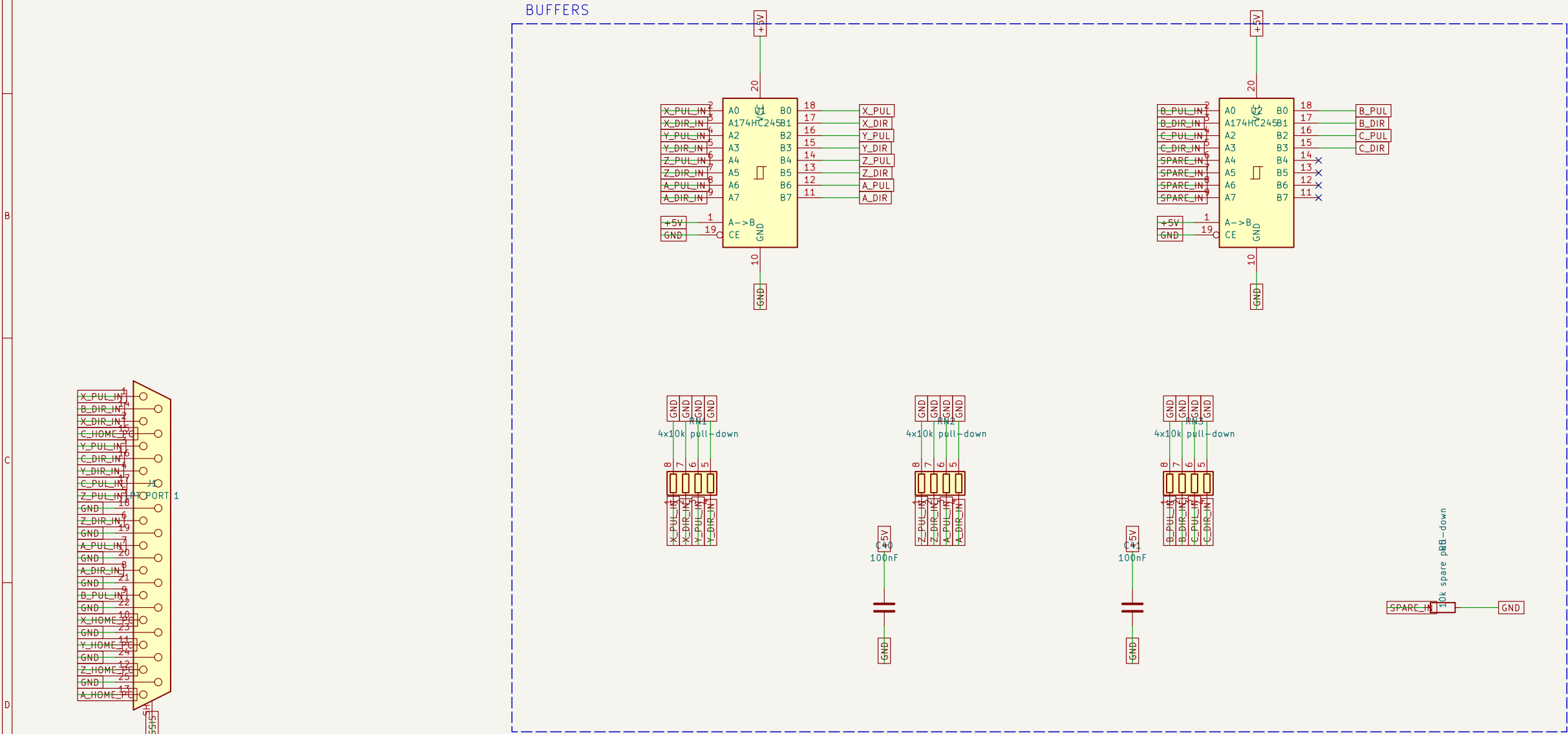


Sheet: /		
File: MACH3SIMPLE.kicad_sch		
Title: MACH3-SIMPLE: copy of the board in service		
Size: A4	Date:	Rev: A1 ENGINEERING ONLY
KiCad E.D.A. 10.0.6		Id: 1/6



LPT PORT 1 – MOTION. Pin numbers are the ones printed on the board in service, so the existing Mach3 profile carries over unchanged.

NOT ISOLATED: the axis terminals share ground with the PC. That is the arrangement on the board in service and the one chosen for this copy.



INPUTS. 24 V sensors on optocouplers, handwheel straight through.

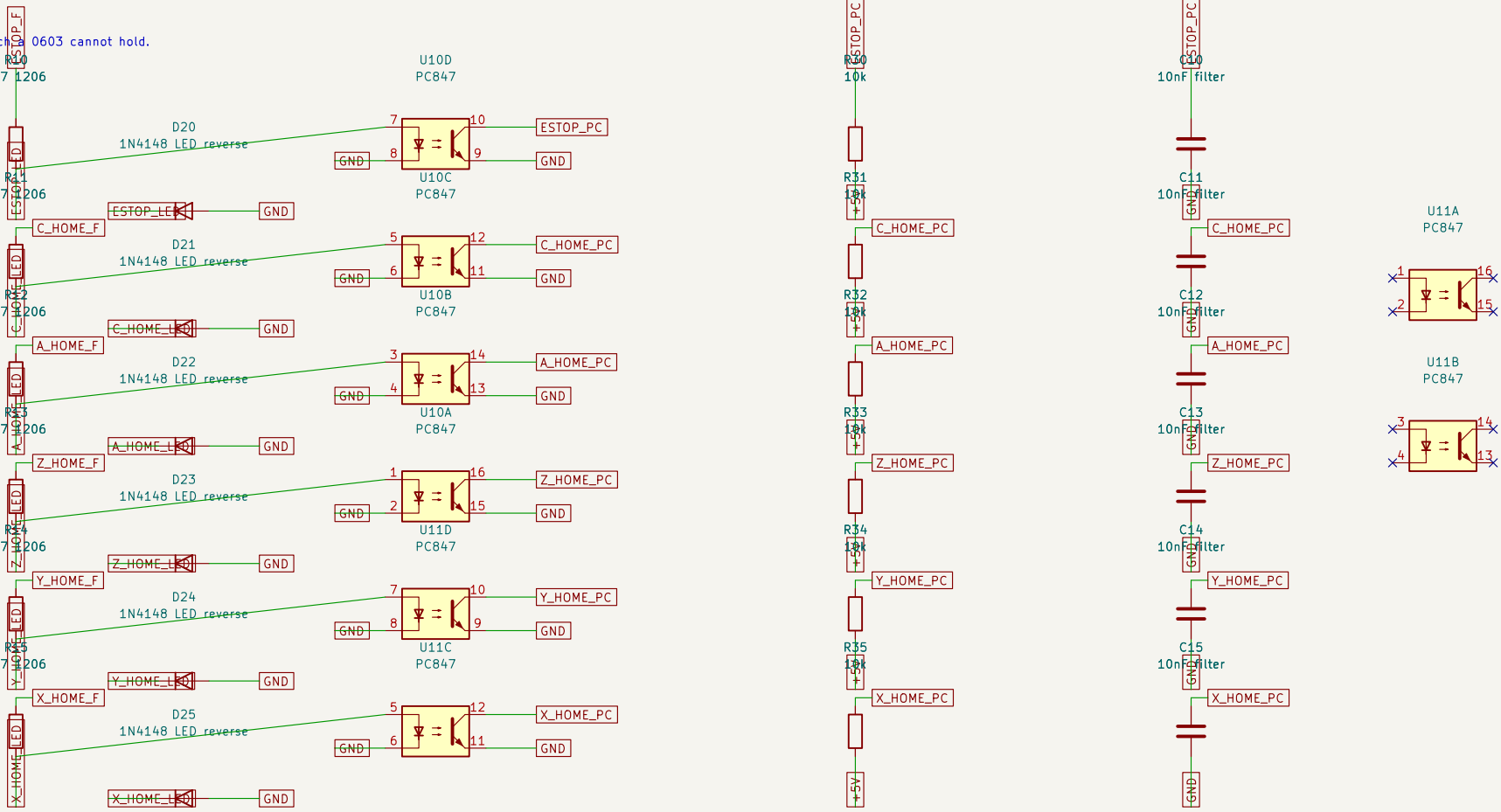
EVERY optocoupler LED gets an anti-parallel diode. A PC847 LED is rated

6 V reverse and a sensor cable beside a motor lead delivers far more.

That is the most likely reason optocouplers keep dying, and the fix is one diode each.

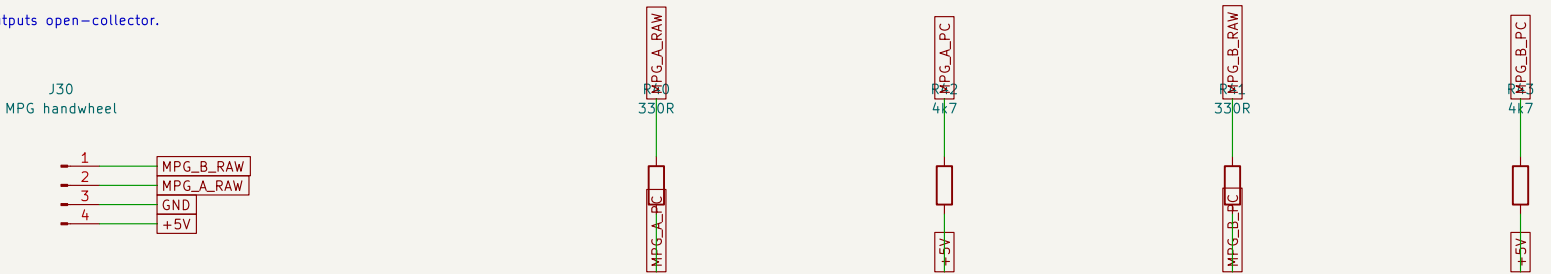
24 V INPUT CHANNELS

4k7 in 1206: $(24-1.2)^2/4700 = 0.11$ W, which 0603 cannot hold.



HANDWHEEL – 5 V, STRAIGHT THROUGH

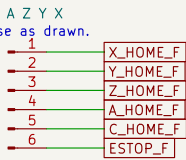
The MPG runs at 5 V and shares ground with the port, so it needs no optocoupler – only a series resistor and a pull-up, because most handwheels drive their outputs open-collector.



EMG & HOME – ONE POSITION PER SIGNAL

One six-way block, labelled EMG C A Z Y X as on the board in service.

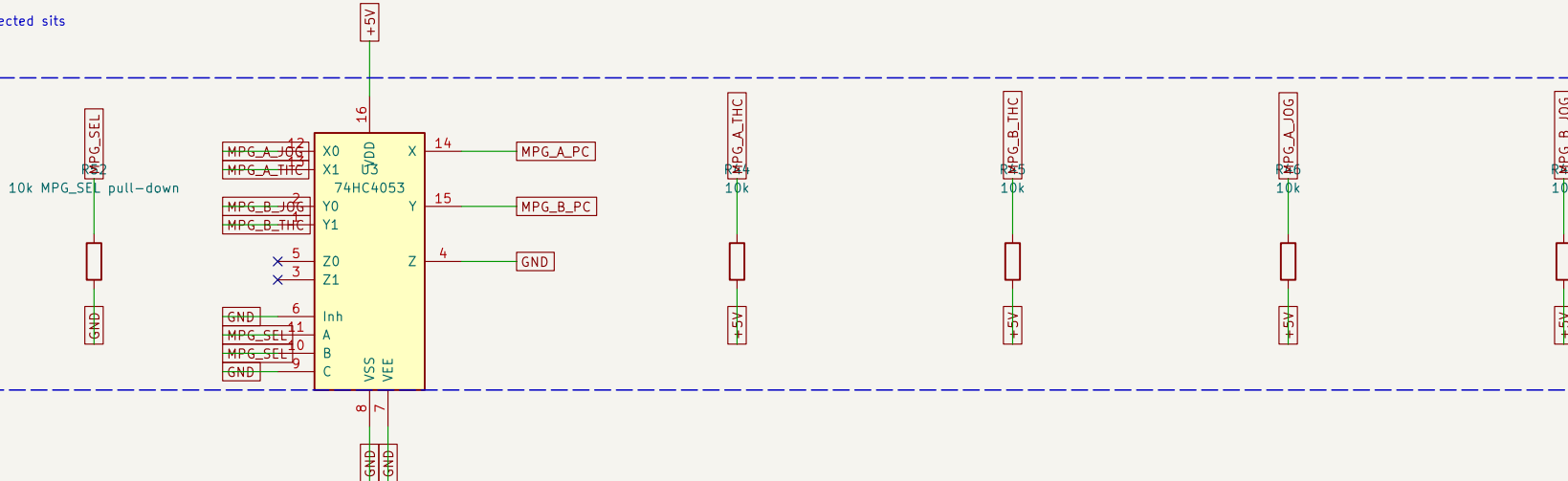
The sensors take their +24 and 0 V from the 24 V terminal, not from a 24V regulator. PNP (sourcing) sensors: an NPN will not drive these as drawn.



HANDWHEEL ROUTING – JOG OR THC, CHOSEN BY SendSerial

One encoder, two jobs. U3 sends MPG A and B either to port 2 pins 12 and 13, where Mach3 reads them as the handwheel, or to pins 10 and 11, where it reads them as THC up and down. The MCU picks, on SendSerial 'T' for THC and 't' for jog, so it is one more macro alongside the relays.

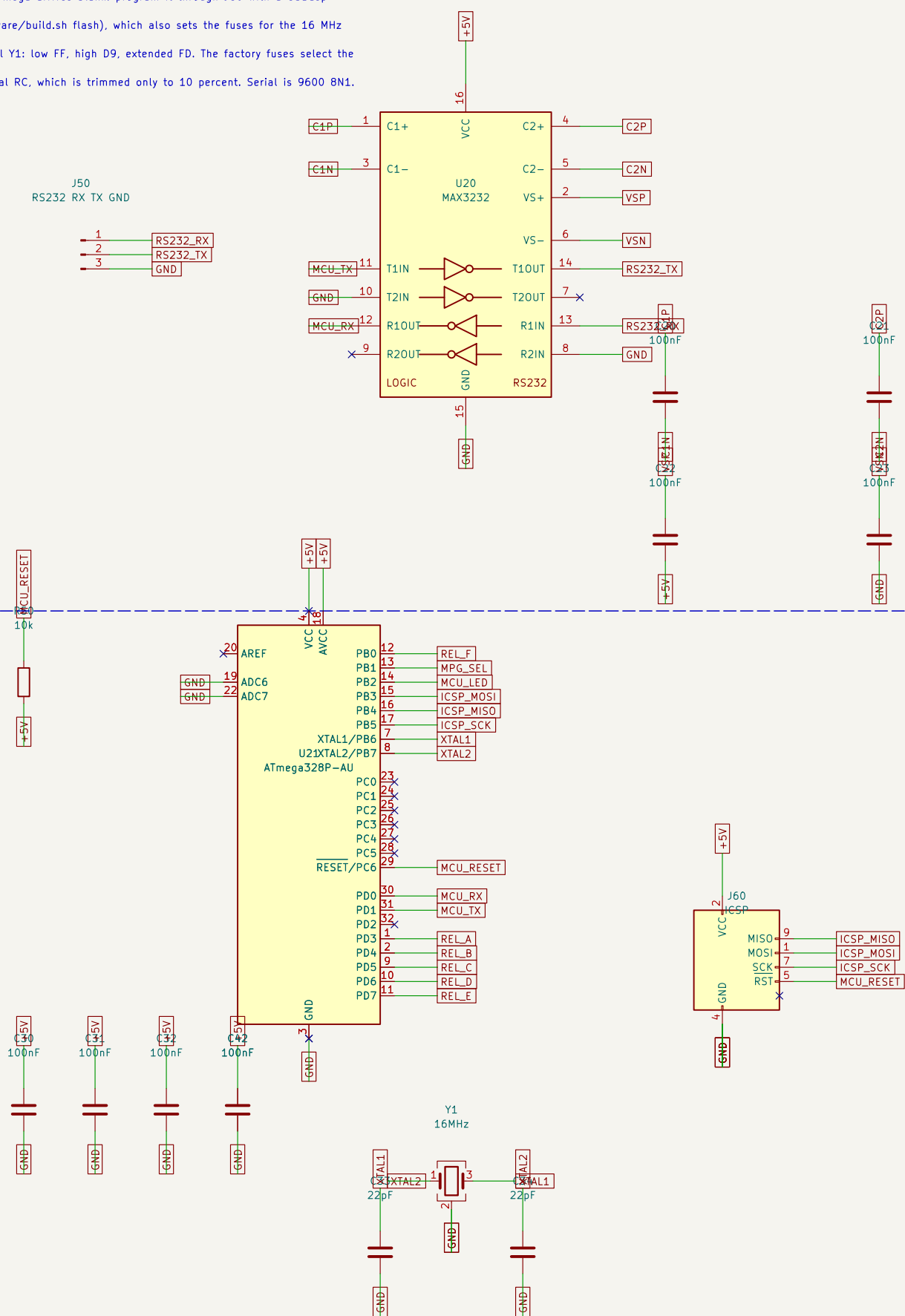
The four LPT pins are pulled up, so whichever pair is not selected sits high and inactive rather than floating.



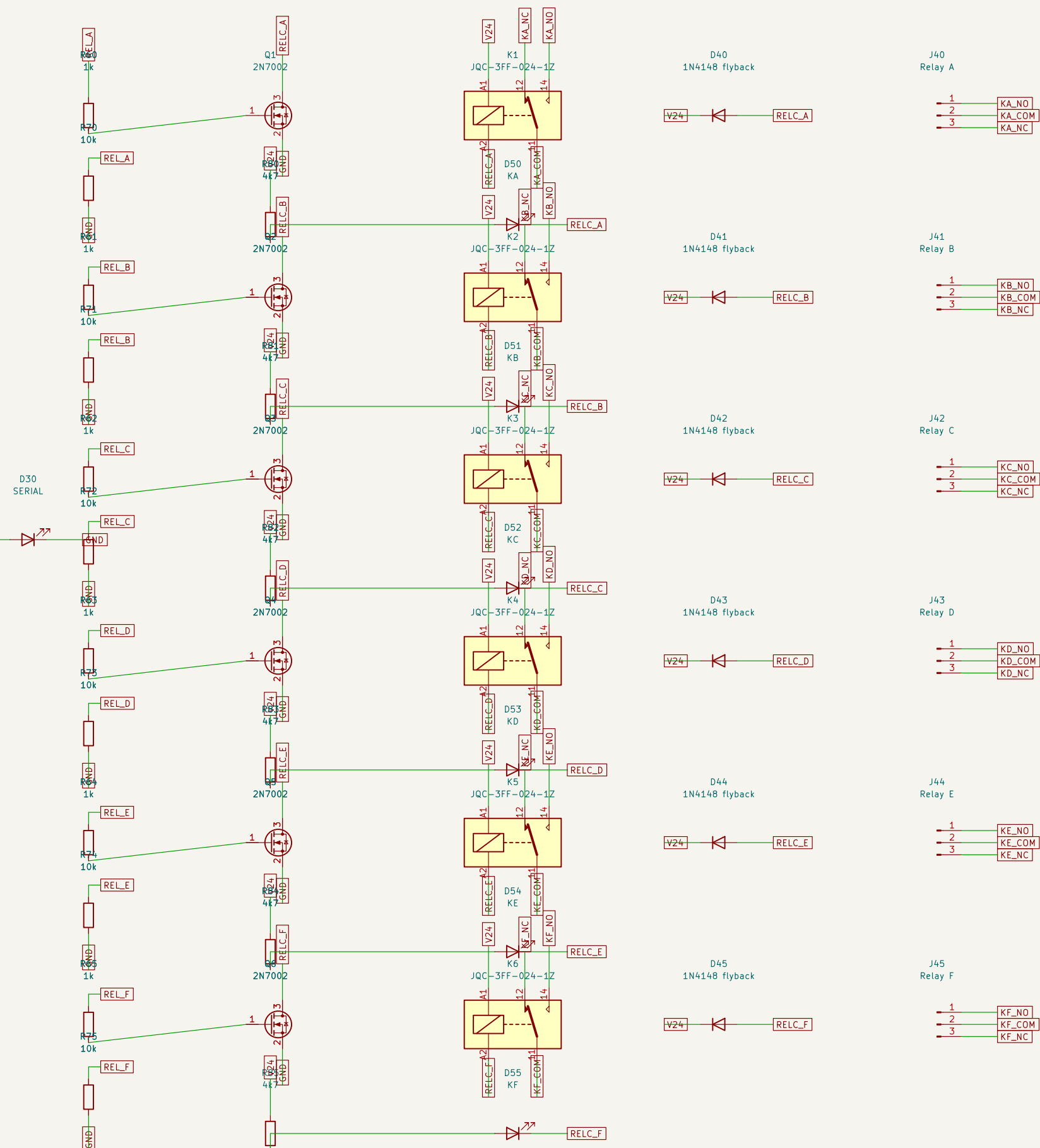
protocol as the board in service, so existing macros keep working.

SIX RELAYS – 24 V COILS ONLY

The ATmega arrives blank: program it through J60 with a USBasp (firmware/build.sh flash), which also sets the fuses for the 16 MHz crystal Y1: low FF, high D9, extended FD. The factory fuses select the internal RC, which is trimmed only to 10 percent. Serial is 9600 8N1.



The contacts switch 24 V Finder coils. Nothing on this board touches mains. A Finder coil is 20–60 mA, so a 10 A contact is heavily derated.



SPINDLE. 0-10 V speed and FWD / REV for the VFD, isolated from this board.

Mach3: port 2 pin 7 = spindle PWM, PWMBase Freq 100 Hz. Pin 8 = Output #1 (M3, FWD).

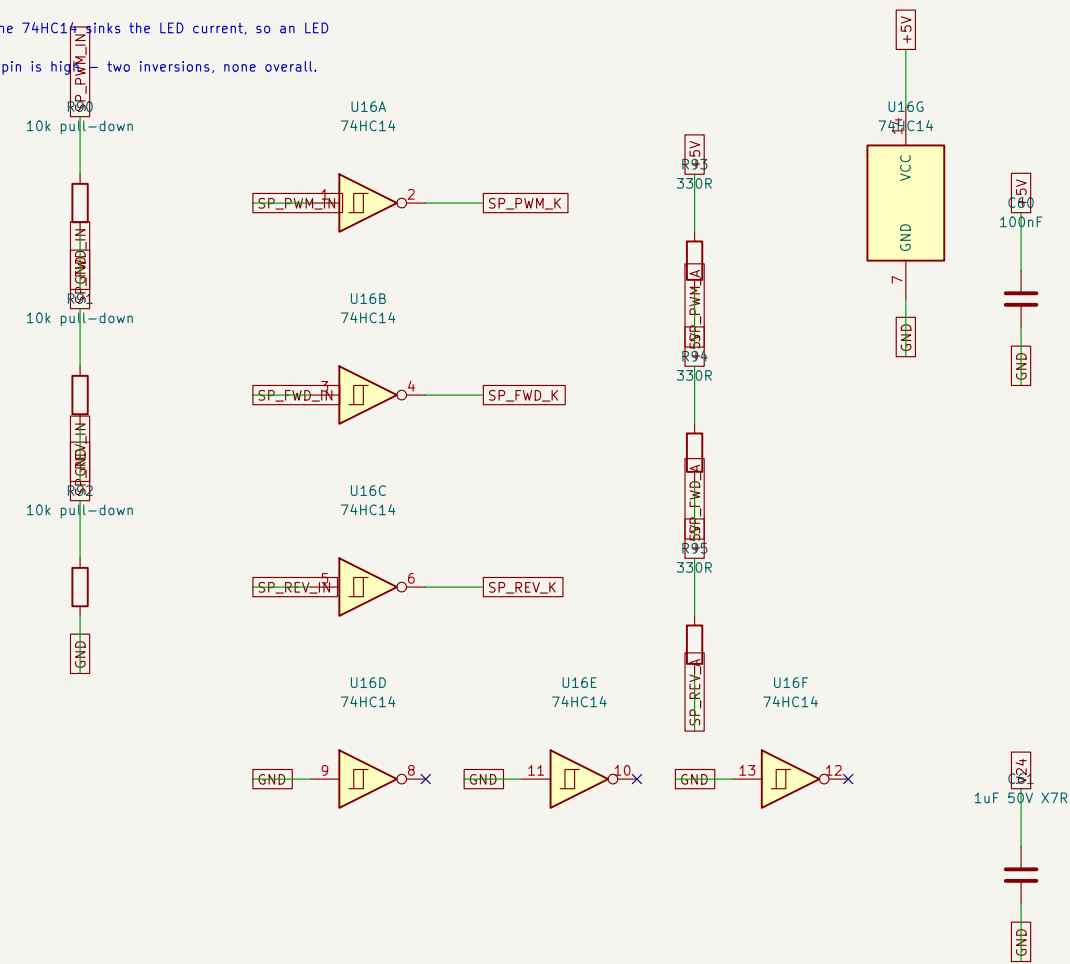
pin 9 = Output #2 (M4, REV). All three active high. Trim RV1 for 10.0 V at full speed.

BOARD SIDE – PORT 2 TO THE LEDS

Pulled down, so with the PC off or the cable out every LED is dark:

no speed, no direction. The 74HC14 sinks the LED current, so an LED

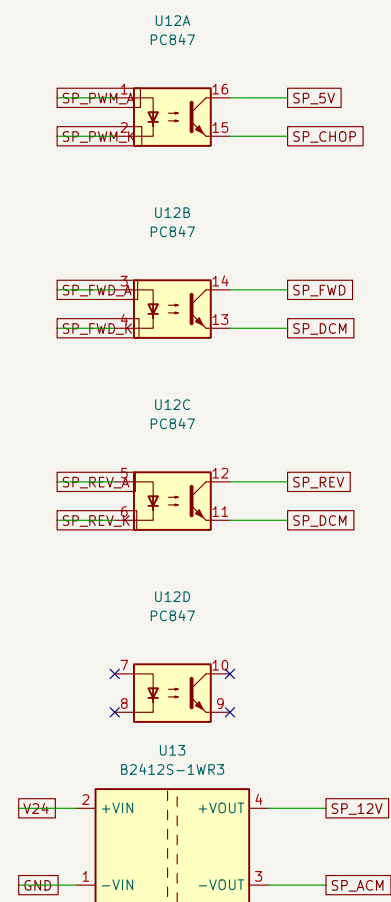
lights only when its port pin is high – two inversions, none overall.



BARRIER

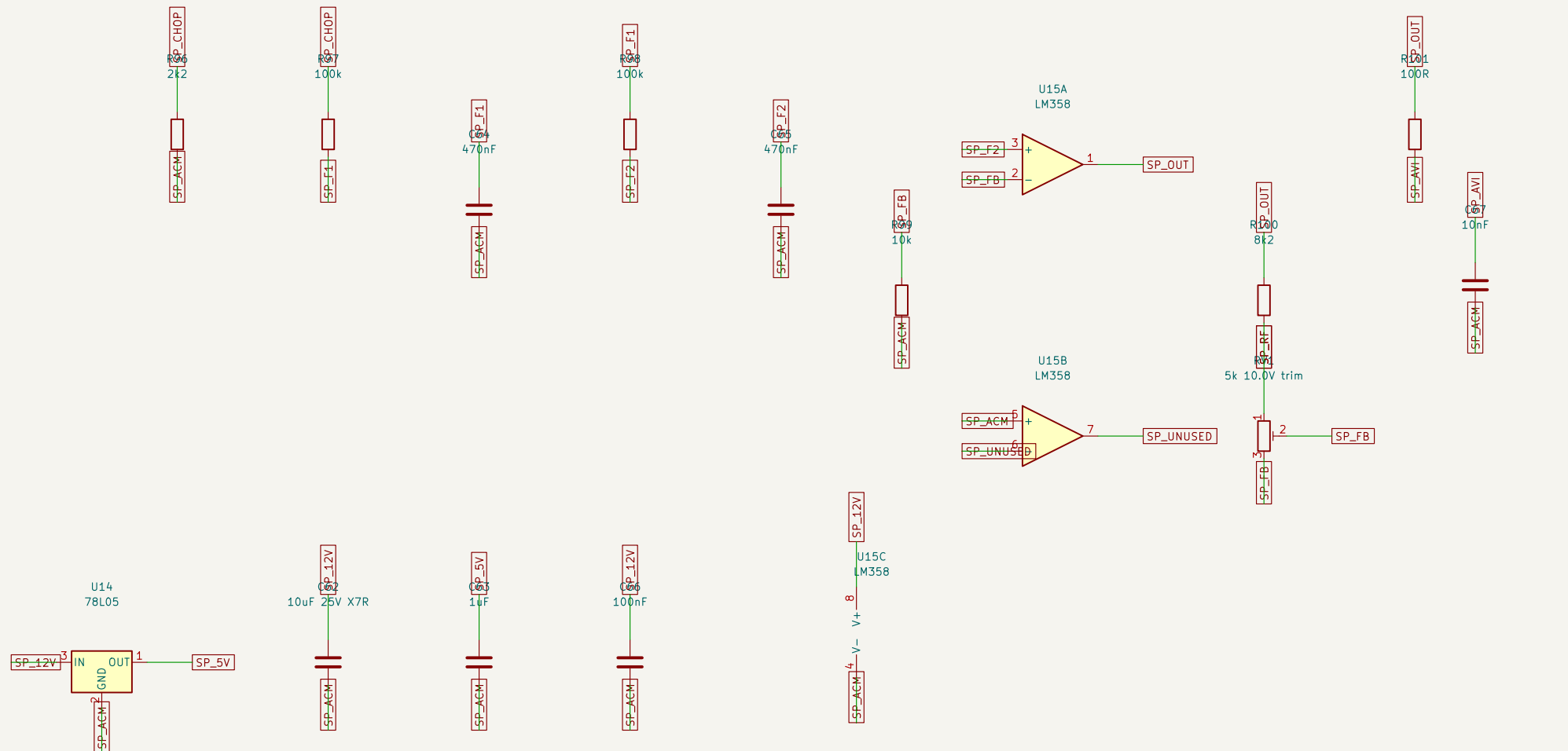
Nothing crosses this box but light

and the DC–DC module.



VFD SIDE – ISOLATED, ITS GROUND IS THE VFD ACM

SP_ACM and SP_DCM are the VFD commons. Neither touches GND.



VFD TERMINAL

AVI ACM FWD REV DCM. Most VFDs ship with FWD/REV in NPN (sink) mode.

which is what these outputs need. PC847: 35 V, 50 mA per output.

